

**INCLUSIVE MOBILE LEARNING APP FOR VISUALLY,
HEARING, PHYSICALLY AND COGNITIVELY
IMPAIRED STUDENTS (GRADE 3-5)**

25-26J-207

Project Proposal Report

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Sri Lanka Institute of Information Technology
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date: 2025/08/28

ABSTRACT

Hearing impairment is one of the most common challenges that affect primary school children in Sri Lanka, where most classroom lessons depend on spoken instructions. Existing international solutions, such as those using American or British Sign Language (ASL) or British Sign Language (BSL), are not suitable for Sri Lankan Sign Language (SLSL) and the Sinhala language. This research proposes a mobile learning application developed with Flutter to support students in Grades 3—5 who are deaf or hard of hearing. The system delivers lessons through SLSL video content with Sinhala text, supported by animations and diagrams to improve comprehension. A camera—based gesture recognition system using pose estimation is integrated to evaluate sign imitation, enabling gesture scoring and interactive sign practice lesson-based games. The application also includes animated story—based explanations for complex topics and a Sinhala—language Chabot that generates quizzes, provides instant feedback, and guides students through lessons. Furthermore, parents and teachers can view progress through detailed performance reports. By combining accessibility, localization, and AI (Artificial Intelligence)—driven interactivity, the proposed application provides an inclusive educational platform tailored to the cultural and curricular needs of Sri Lankan hearing-impaired students.

Keywords: Hearing impairment, Sri Lankan Sign Language (SLSL), Mobile learning application, Gesture recognition, Inclusive education

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LIST OF ABBREVIATIONS

Abbreviation	Description
SL	Sign Language
AI	Artificial Intelligence
ASL	American Sign Language
BSL	British Sign Language
SLSL	Sinhala Sign Language
SLR	Sign Language Recognition
GRU	Gated Recurrent Unit
1D-CNN	One-Dimensional Convolutional Neural Network
DTW	Dynamic Time Warping
TFLite	TensorFlow Lite
CNN	Convolutional Neural Network
LLM	Language Model

SVM	Support Vector Machine
LSTM	Long Short-Term Memory
OCR	Optical Character Recognition
ML	Machine Learning
IP	Image Processing
HCI	Human–Computer Interaction

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1. INTRODUCTION

1.1 BACKGROUND

Hearing impairment is a significant barrier to learning in traditional classroom environments, particularly for Sri Lankan students in Grades 3 to 5. Most educational materials are designed with spoken instructions, making it difficult for deaf and hard-of-hearing children to follow lessons. This situation limits both academic progress and social inclusion, creating an urgent need for accessible, localized learning solutions.

Global efforts have introduced Sign Language (SL) videos, captioning, interactive multimedia, and Artificial Intelligence (AI) powered accessibility tools. These systems demonstrate the value of visual and text-based learning approaches but are largely developed for American Sign Language (ASL), British Sign Language (BSL), or other well-resourced languages. Few systems are tailored to Sinhala Sign Language (SLSL), which reduces their effectiveness for Sri Lankan students.

Modern Sign Language Recognition (SLR) systems rely on two main components: pose/key-point extraction and gesture recognition models. Frameworks such as MediaPipe Hands and BlazePose enable real-time landmark extraction of hands and body, making them suitable for mobile deployment in low-resource school environments. Building on this, deep learning models such as Gated Recurrent Unit (GRU), One-Dimensional Convolutional Neural Network (1D-CNN), and Dynamic Time Warping (DTW) can be applied to evaluate gestures. Unlike existing Sinhala-based systems that use binary classification (correct/incorrect), this approach allows detailed numeric gesture scoring.

The proposed solution introduces novelty through sign-imitation lesson-based games aligned with the national curriculum. Learners practice SLSL signs through imitation activities, receive personalized calibration based on their speed and style, and obtain real-time feedback with motivational scores. Gamified quizzes, points, and badges further

enhance engagement, while progress tracking dashboards enables teachers and parents to monitor performance and provide additional support.

Deployment on Flutter with TensorFlow Lite (TFLite) ensures real-time processing on mobile devices, while Firebase and MongoDB provide secure data storage and quiz backend management. This combination of localized SLSL lessons, pose-based scoring, AI-driven feedback, and interactive gamification fills a clear gap in existing research and applications. By directly addressing the needs of Sri Lankan Grade 3–5 students, the system creates a culturally relevant, accessible, and scalable platform for inclusive education.

1.2 LITERATURE REVIEW

Gesture based Sign Language Recognition system using Mediapipe [1].

This system demonstrated that lightweight vision-based pipelines can be efficient for gesture classification without requiring specialized hardware like gloves or Leap Motion. However, its primary focus remained recognition-only tasks—identifying signs correctly—without extending into educational applications such as gesture scoring, sign imitation, or gamification. In contrast, the proposed system advances beyond recognition by incorporating DTW, GRU, and 1D-CNN to provide numeric gesture scoring. It also introduces lesson-based sign imitation games aligned to the Sri Lankan Grade 3–5 curriculum, supported by progress tracking and personalized calibration. While the Mediapipe + Support Vector Machine (SVM) approach validates technical feasibility for recognition, the proposed system leverages similar vision pipelines but transforms them into an interactive learning environment specifically designed for SLSL, filling both cultural and educational gaps.

Sinhala Sign Language Recognition using Leap Motion and Deep Learning [2].

This study used a Leap Motion sensor and Convolutional Neural Networks (CNNs) to classify SLSL gestures. The hardware provided skeletal hand data with good precision, but the system was limited to a binary classification model, identifying signs only as correct or incorrect. Furthermore, the reliance on external hardware makes it unsuitable for wide adoption in schools where cost and accessibility are barriers. The proposed system eliminates this dependency by using camera-based pose estimation through MediaPipe Hands and BlazePose, offering a more scalable solution. More importantly, it moves beyond binary classification to numeric gesture scoring and integrates curriculum-based imitation games, directly addressing the shortcomings of the Leap Motion approach.

Enhancing the Communication Experience for the Deaf and Hard-of-Hearing using AI Language Models [3].

This work explored AI Language Models (LLMs) for bridging communication between deaf and hearing individuals. While impactful in enabling conversations, its focus was on general communication support rather than structured educational learning. It lacked sign imitation, scoring, or SLSL-specific implementation. The proposed system differs by addressing education rather than conversation, offering lesson-based imitation games and performance evaluation aligned with the Sri Lankan primary school curriculum. Thus, while LLMs serve as conversational aids, they do not meet the educational needs of hearing-impaired children in Sri Lanka, a gap the proposed solution targets.

A Comprehensive Review of Sign Language Recognition: Different Types, Modalities, and Datasets [4].

This survey highlighted sensor-based (gloves, Leap Motion) and vision-based (CNNs, Mediapipe) approaches for Sign Language Recognition (SLR), along with dataset limitations. However, the work was descriptive only, without presenting a usable application for learners. The proposed system directly applies the reviewed technologies, particularly pose estimation with GRU/1D-CNN, to create a real-time educational tool.

By embedding these methods into curriculum-linked lesson games, the system extends beyond recognition into active learning and feedback, which was absent in the review.

Sinhala Sign Language Learning System for Hearing Impaired Community [5].

This system targeted the Sinhala community with a recognition-based learning application. While relevant to the local context, its design offered limited interactivity, only recognizing whether a sign was correct. It lacked quantitative evaluation and engagement features such as imitation games. The proposed system improves upon this by incorporating numeric scoring, personalized calibration, and gamified sign imitation, ensuring learning is both interactive and measurable.

Hastha: Online Learning Platform for Hearing Impaired [6].

Hastha provided an accessible online platform with lessons and quizzes for hearing-impaired students. However, the learning model was passive, focusing on multiple-choice questions and static quizzes. It did not employ real-time imitation or pose-based evaluation. In comparison, the proposed system shifts from passive testing to active, camera-based imitation games, enabling children to practice signs in real time, receive scoring-based feedback, and track their progress.

SLRNet: A Real-Time LSTM-Based Sign Language Recognition System [7].

SLRNet demonstrated real-time sign recognition using Long Short-Term Memory (LSTM) networks. While technically advanced, it focused primarily on ASL datasets and recognition performance. There was no attempt to provide localized educational content for Sri Lankan learners or to integrate gamification. The proposed system fills this gap by focusing on SLSL for Grades 3–5, combining recognition accuracy with lesson-based games and scoring to enhance classroom engagement.

Enhancing Sign Language Detection through Mediapipe and Convolutional Neural Networks (CNN) [8].

This study validated Mediapipe and CNN as effective tools for gesture detection. However, it remained a detection-focused research experiment and did not consider education, gamification, or localization. The proposed system extends these findings into a practical educational application, introducing imitation games, numeric scoring, and curriculum-aligned learning for SLSL. Thus, it transforms technical validation into real-world use for learners.

Evaluating the Immediate Applicability of Pose Estimation for Sign Language Recognition [9].

This benchmarking study assessed pose estimation frameworks like OpenPose and Mediapipe for SLR feasibility. While it showed promise, it lacked scoring mechanisms, imitation games, and local adaptation. The proposed system addresses these gaps by operationalizing pose estimation into a learning app, embedding imitation-based quizzes, numeric scoring, and progress monitoring within a localized SLSL educational framework.

A Multifunctional Communication System for Differently Abled People [10].

This system combined features such as Optical Character Recognition (OCR), speech support, and gesture recognition. Its broad focus supported multiple disabilities but was not designed for SLSL learning or imitation scoring. The proposed system narrows the focus specifically to hearing impairment, providing a dedicated SLSL education tool with gamification and measurable performance feedback, which the multifunctional system lacked.

“BeMyVoice” – A Communicative Platform to Assist Deaf Students [11].

BeMyVoice was designed for communication in classrooms, including basic gamification with quizzes. However, it did not integrate imitation-based feedback or quantitative scoring. The proposed system builds upon the gamification idea but makes it curriculum-aligned and measurable, offering sign imitation games that directly improve learning outcomes for children.

EasyTalk: A Translator for Sri Lankan Sign Language using Machine Learning and Artificial Intelligence [12].

EasyTalk focused on accessibility by creating a sign-to-text translator for SLSL. While useful for communication, it did not address education or learning reinforcement. The proposed system transforms the focus from translation to education, using sign imitation, scoring, and gamification to help children actively develop signing skills aligned with the curriculum.

E-Learning Platform for Hearing Impaired Students [13].

This platform delivered structured e-learning modules and quizzes but did not adopt imitation or scoring mechanisms. Learning remained traditional. The proposed system enhances e-learning by enabling real-time imitation practice, personalized scoring, and engagement through badges and rewards, making learning more interactive and child-centered.

Empowering Deaf Children with Sinhala Sign Language, Emotion Detection, and Sound Recognition [14].

This work integrated emotion detection and sound recognition with SLSL for a multimodal support system. While innovative, its focus was not sign imitation games or performance scoring. The proposed system provides this missing dimension by offering lesson-based imitation games, scoring feedback, and progress tracking, ensuring measurable skill acquisition alongside engagement.

Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka [15].

Visual Kids provided a child-friendly app with interactive activities but relied on general learning games without pose estimation or imitation scoring. The proposed system builds on its engaging design but adds numeric scoring, personalized calibration, and curriculum-linked imitation games, making it more educationally effective.

Utalk: Sri Lankan Sign Language Converter Mobile App using Image Processing and Machine Learning [16].

Utalk developed a Machine Learning (ML)-based Image Processing (IP) translator for SLSL. Its goal was conversion, not education, and it lacked gamification or scoring. The proposed system redefines this approach by shifting from translation to teaching, allowing children to imitate signs, receive scoring-based feedback, and practice through gamified lessons.

HANDTALK: Adaptive and Interactive Self Learning System for Deaf and Dumb School Students in Sri Lanka [17].

HANDTALK promoted adaptive self-learning but did not include quantitative imitation scoring or gamified sign practice. It supported progression but lacked measurable performance indicators. The proposed system integrates pose-based scoring and gamified imitation games, ensuring learners not only progress but also measure their accuracy levels in real time.

Smart System to Support Hearing Impaired Students in Tamil [18].

This Tamil system incorporated multimodal features like sign↔text, speech, lip reading, and object recognition, showing innovation in accessibility. However, it was not an educational imitation system and did not include scoring. The proposed system addresses this by providing a Sinhala-specific educational app with numeric scoring, imitation-based games, and curriculum integration, focusing on teaching rather than translation.

Sinhala Sign Language Translation through Immersive 3D Avatars and Adaptive Learning [19].

This system used 3D avatars with adaptive learning features. While it gamified lessons, the games were quiz-oriented and lacked imitation-based scoring. The proposed system introduces sign imitation games with numeric gesture scoring, allowing learners to receive measurable feedback that avatars alone could not provide.

IHear- Mobile Application for Hearing- Impaired Children to Facilitate Their Hearing and Speech Abilities [20].

IHear focused on hearing and speech rehabilitation, offering speech scoring and sound classification. While valuable, it was not sign-based and did not support SLSL education. The proposed system translates the scoring methodology into sign language learning, with pose-based gesture scoring and lesson-based gamification for primary learners, filling a critical educational gap.

1.3 RESEARCH GAP

Hearing impairment is one of the most significant barriers to inclusive education, particularly for primary school children who rely heavily on spoken instruction. In Sri Lanka, students in Grades 3–5 who are deaf or hard of hearing face difficulties following mainstream lessons, as most educational content is designed around verbal delivery and auditory cues. While global research has explored various assistive technologies such as sign language recognition, captioning, and multimedia integration, very few systems have been adapted to the local context of SLSL and the Sri Lankan school curriculum. This mismatch between existing global solutions and local needs creates a clear research gap that limits accessibility and equal learning opportunities for Sri Lankan students.

A major shortcoming in prior research is the dominance of systems built for foreign sign languages such as ASL or BSL. These solutions, while technically advanced, cannot be directly applied in Sri Lanka because of linguistic and cultural differences in sign usage. Even when attempts have been made to develop tools for Sinhala Sign Language, they have mostly been limited to recognition tasks that classify signs as either correct or incorrect. This binary approach is insufficient for educational purposes, as it fails to give learners detailed feedback about the quality of their signing or how to improve. For young learners, particularly those in primary school, such detailed and formative feedback is crucial for building confidence and improving accuracy over time.

Another significant gap lies in the absence of gamified, sign-imitation-based learning approaches. Most existing systems use static resources such as video libraries, multiple-choice quizzes, or basic translation tools. These methods are limited in their ability to engage children, as they lack interactive features that make learning enjoyable and motivating. Research shows that gamification, when combined with real-time feedback, increases learner engagement and retention. However, current Sinhala Sign Language applications rarely integrate interactive imitation games where children can practice copying signs and receive measurable feedback. Without these elements, learners may lose motivation, and the educational experience becomes passive rather than active.

Previous studies have also highlighted the use of external hardware, such as Leap Motion controllers, gloves, or depth sensors, for gesture recognition. While these devices can capture fine-grained motion data, they are expensive and impractical for deployment in low-resource classrooms or rural Sri Lankan schools. Reliance on specialized hardware creates a barrier to scalability, as most schools only have access to smartphones or basic computing devices. This makes many of the existing foreign solutions unsuitable for the Sri Lankan education sector, where affordability and accessibility are essential.

Even in cases where recognition technologies have been applied, there are further gaps in personalization and feedback. Most systems do not adapt to a child's hand size, signing style, or speed. As a result, learners may receive inaccurate evaluations that discourage them rather than support improvement. Personalized calibration, where the system adjusts scoring to match each learner's rhythm and physical characteristics, is largely missing from existing applications. This gap is especially critical for children in the developmental learning stage, who require tailored feedback to progress effectively.

Equally important is the limited integration of progress monitoring and reporting features in existing platforms. Parents and teachers need actionable insights into a student's learning journey to provide appropriate support. However, most current systems stop at recognition or quiz-based assessments without offering detailed progress reports. Without structured tracking, it becomes difficult for teachers to identify areas where students struggle, and for parents to actively participate in their child's learning process. Thus, the

absence of progress dashboards represents another critical gap that reduces the long-term educational value of these systems.

From a technical perspective, many of the surveyed works stop short of combining recognition with real-time feedback, gamified learning, and curriculum alignment. Benchmark studies using pose estimation frameworks such as MediaPipe or OpenPose demonstrate that vision-based key point extraction can effectively identify signs. However, these studies remain at the level of recognition accuracy testing rather than educational application. They do not extend into building end-to-end learning environments that incorporate practice, feedback, and assessment. This limits their applicability for primary education, where the goal is not only recognition but also sustained skill development through interactive practice.

Finally, when analyzing the collective body of work, it becomes evident that no existing system brings together the full set of features required for a comprehensive, localized learning platform. Specifically, there is no single mobile application that integrates Sinhala Sign Language video lessons, real-time camera-based gesture recognition with numeric scoring, curriculum-aligned quizzes, animated visual aids, gamified imitation games, and progress tracking for teachers and parents. Each of these features has been addressed in isolation by different research projects but never combined into one cohesive system. This fragmentation leaves a significant research and implementation gap that directly affects the accessibility and inclusiveness of education for hearing-impaired children in Sri Lanka.

In summary, the research gap can be articulated across several dimensions: lack of focus on Sinhala Sign Language and the local curriculum, overreliance on binary recognition models without detailed scoring, absence of sign-imitation gamification, dependence on expensive hardware, lack of personalization and real-time feedback, and poor integration of progress monitoring tools. Bridging this gap requires an all-in-one, child-friendly mobile application that combines pose-based recognition, quantitative scoring, gamification, localized content, and progress analytics within a deployable and low-cost architecture. Addressing these gaps will ensure that hearing-impaired students in Sri

Lanka gain equal opportunities for interactive, measurable, and culturally relevant education.

Table I.: Comparison of Research Gaps in Existing Studies and Proposed Mobile Application

Feature Research Paper	Sri Lankan Sign Language (SLSL) Support-Sinhala	Gesture Scoring	Sign Imitation Lesson Based Games	Games	Quizzes
Gesture based Sign Language Recognition system using Mediapipe	(ASL/ISL not SLSL) 				
Hastha: Online Learning Platform for Hearing Impaired				(text/matching-type games) 	
Smart System to Support Hearing Impaired Students in Tamil	(Tamil SL) 				
Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka				(matching, typing, and text-based learning games) 	
E-Learning Platform for Hearing Impaired Students	(ASL) 				
Our Proposed System					

1.4 RESEARCH PROBLEM

In Sri Lanka, many Grade 3–5 students who are deaf or hard of hearing struggle to access education due to spoken instruction–based teaching methods that create a communication barrier. International tools designed for ASL or BSL are not effective for local learners, as they do not match the grammar, vocabulary, or culture of SLSL. Existing Sinhala-based tools are limited to video lessons and subtitles, with no interactive features such as gesture scoring, gamification, or AI-driven feedback. Moreover, most systems are not aligned with the Sri Lankan curriculum, which reduces their usefulness in classroom settings. This gap highlights the need for a localized, curriculum-aligned mobile learning application that integrates SLSL lessons, numeric gesture scoring, sign-imitation games, visual storytelling, AI-based quizzes, and progress tracking to provide an inclusive and engaging learning experience.

1. Why do hearing-impaired students in Grades 3–5 face barriers in Sri Lankan classrooms?

Most lessons depend on spoken instructions and teacher explanations, which deaf students cannot fully access, causing knowledge gaps and slower academic progress.

The system introduces SLSL-based video lessons with Sinhala subtitles and visuals, ensuring content is accessible without relying on spoken language.

2. Why are international ASL/BSL-based solutions not suitable for Sri Lankan learners?

They are designed for foreign sign languages, with different grammar and vocabulary, making them ineffective for students who use SLSL. The mobile app is built specifically for SLSL and is aligned with the local Grade 3–5 curriculum, making it culturally and linguistically relevant.

3. What limitations exist in current Sinhala-based educational tools?

They provide only basic video lessons or subtitles without gesture scoring, AI feedback, or interactive sign-imitation features, limiting active participation. The system integrates pose-estimation–based gesture scoring with AI-driven feedback, and adds sign-imitation games, making learning interactive and measurable.

4. How does the lack of curriculum alignment affect students?

Most tools are not mapped to the Sri Lankan Grade 3–5 syllabus, so learners miss locally relevant topics, cultural terms, and examples, making learning less relatable.

The proposed system delivers curriculum-linked lessons, quizzes, and games specifically designed for the Sri Lankan context, ensuring direct classroom integration.

5. What solution can you overcome these challenges?

A localized mobile application with SLSL support, numeric gesture scoring, sign-imitation games, AI-driven quizzes and progress tracking can provide and engaging and curriculum-aligned learning experiences for deaf students in Sri Lanka.

The system combines SLSL lessons, AI-based quizzes, gamification and teacher/parent dashboards to create a complete, interactive learning platform for hearing-impaired students.

2. OBJECTIVES

2.1 MAIN OBJECTIVES

To design and develop a mobile learning application using SLSL, AI-powered sign recognition, and interactive visual tools that provide an accessible, engaging, and curriculum-aligned learning experience for Sri Lankan primary students in Grades 3–5 who are deaf or hard of hearing.

2.2 SPECIFIC OBJECTIVES

1. Deliver lesson content in SLSL video format with clear Sinhala subtitles and supporting diagrams, ensuring all key concepts are fully understandable without relying on audio.
2. Implement camera-based sign recognition using pose estimation to detect and interpret students' gestures in real time.
3. Develop gesture scoring and sign imitation games to make sign language learning interactive, motivating, and measurable.
4. Integrate a text-based AI chatbot to provide explanations, answer questions, and guide learners through lessons in Sinhala.
5. Generate AI-powered quizzes after each lesson with instant feedback, scores, badges, and progress summaries to encourage continuous learning.
6. Provide animated visual explanations in Sinhala to simplify complex topics using culturally familiar stories and examples.
7. Enable parents and teachers to access progress reports, complete lessons, and quiz performance for better educational support.
8. Maintain data privacy and ethical standards with secure storage, anonymization, and parental consent for child data usage.

3. METHODOLOGY

The proposed research aims to design and develop a mobile application to support hearing-impaired students in Grades 3–5 by delivering curriculum-aligned lessons, quizzes, and interactive activities in SLSL. The system addresses barriers created by spoken-only instructions by offering lessons in visual and text-based formats with SLSL videos, Sinhala subtitles, diagrams, and animations. This ensures that learners can engage effectively without depending on auditory input.

The application will be developed using Flutter, enabling cross-platform support for both Android and iOS devices. To enhance interactivity, the system integrates AI-powered gesture recognition for SLSL practice, sign-imitation games for active learning, and an AI chatbot to provide explanations, answer student queries, and generate quizzes. The methodology follows an iterative development lifecycle, with continuous testing and refinements at each stage to improve usability and effectiveness.

3.1 System Overview

The proposed inclusive mobile learning system adopts a client–server architecture, ensuring accessibility, scalability, and efficient management of resources. The design focuses on supporting students with visual, hearing, physical, and cognitive impairments, while also enabling teachers and parents to monitor progress and provide timely support.

At the front-end, the system is developed using Flutter, ensuring cross-platform compatibility for both Android and iOS devices. The mobile client handles user interaction, sign language video playback, animations, quiz generation, and progress tracking. It also provides multiple control modes such as voice-based, sign-based, text-based, visual/touch input, and dwell-based gesture control, allowing students with different impairments to interact seamlessly.

The backend is powered by Node.js for application logic and MongoDB for managing user data, quiz records, and progress information. AI models for gesture recognition and scoring are integrated into the backend for real-time feedback. Secondary storage is

managed through AWS S3 buckets, which store large datasets, recorded lessons, and analytics securely.

For the Hearing Impairment Support module, the system integrates pose estimation (MediaPipe Hands + BlazePose) combined with GRU/1D-CNN + DTW models for gesture scoring. Students engage in lesson-based sign imitation games, receive real-time feedback with numeric scores, and complete AI-generated quizzes with visual rewards. Teachers and parents can track performance through dashboards that display completed lessons, quiz results, and areas for improvement.

The overall design ensures the application is curriculum-aligned, Sinhala-first, and inclusive, addressing accessibility challenges while maintaining high usability and engagement for Grade 3–5 learners.

Figure 1 illustrates the system overview diagram of the complete inclusive mobile learning application, showing the interaction between students, parents/teachers, and the platform across all disability support modules.

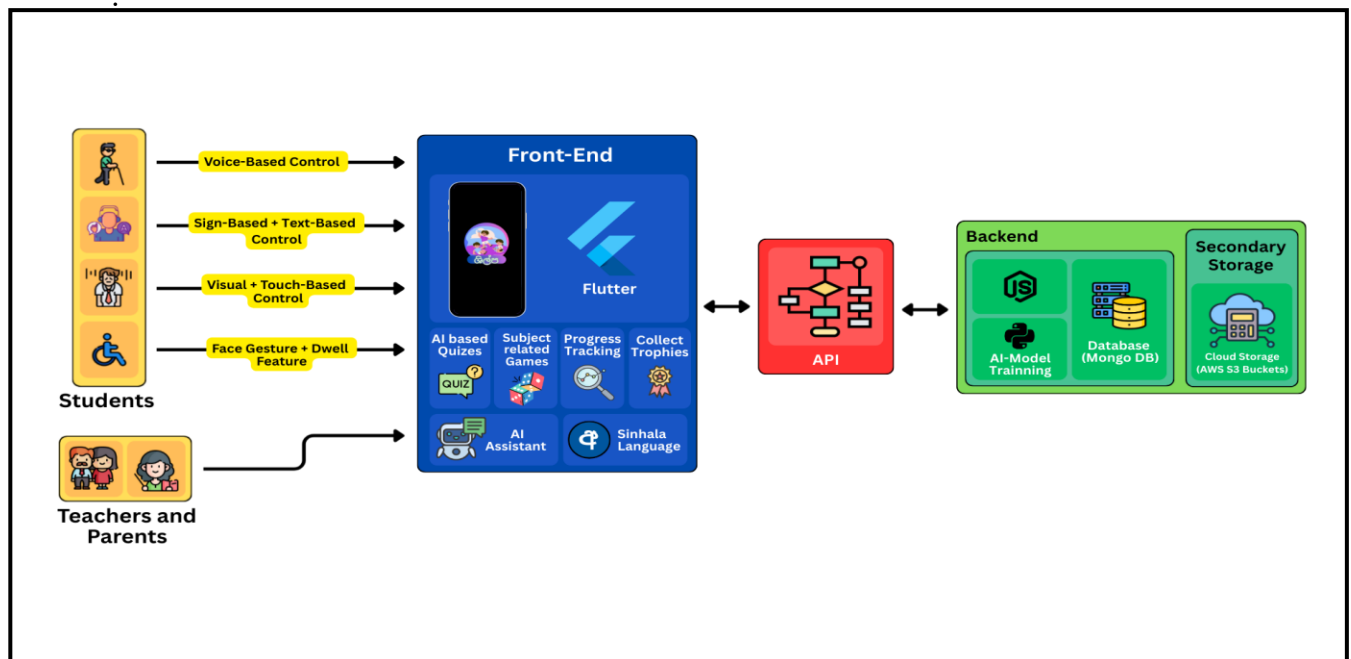


Figure 1: System overview diagram of the complete inclusive mobile learning app

Figure 2 presents the system architecture specifically for the hearing impairment support module, emphasizing how SLSL-based lessons, sign imitation games, pose-estimation models, and AI-powered feedback integrate into the overall application.

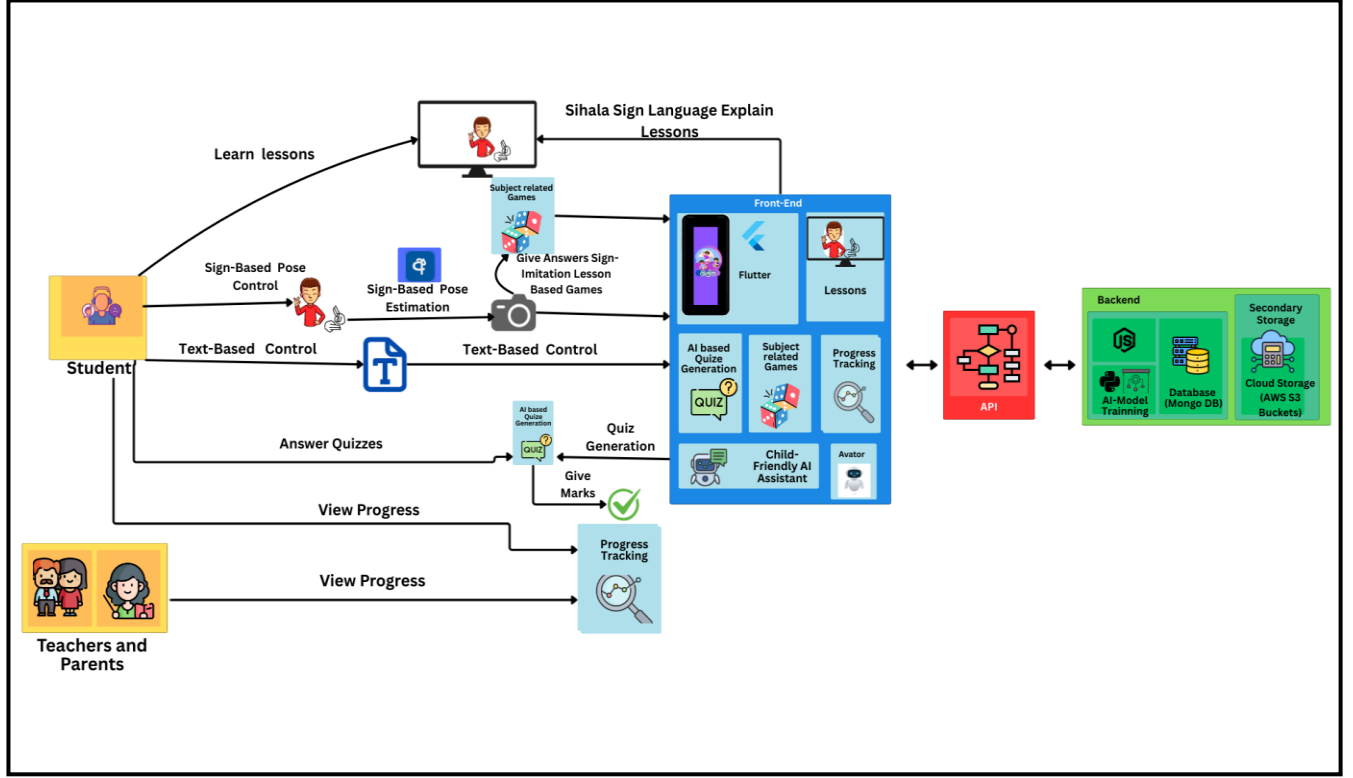


Figure 2: System architecture overview for hearing impairment support

3.2 Development Tools and Technologies

The Hearing Impairment Support application will be developed using a blend of modern tools and specialized domains to ensure accessibility, scalability, and efficiency. Flutter will serve as the cross-platform framework, enabling a single codebase to support both Android and iOS. Backend processing and API development will be managed through Node.js, with MongoDB for storing user progress, quizzes, and reports. Cloud storage (AWS S3/Google Cloud) will be used to host large SLSL video lessons and animations.

For AI and ML, frameworks such as MediaPipe and BlazePose will be integrated for pose estimation, while models like GRU, 1D-CNN, and DTW will be applied for gesture

scoring (0–100). TFLite will allow these models to run efficiently on mobile devices without relying heavily on server resources.

To support Human–Computer Interaction (HCI) and Accessibility, the interface will be designed with Figma to ensure child-friendly visuals, large buttons, and simple navigation tailored for hearing-impaired learners. Inclusive interaction modes such as text-based input, sign-based control, and real-time feedback will be prioritized.

For Data and Ethics, parental consent and child-safe design guidelines will be followed, with anonymized data storage to protect privacy. A Sinhala-enabled chatbot, built with Dialogflow or Rasa, will assist learners in navigating lessons, answering questions, and generating quizzes.

This integration of development tools and domain-specific knowledge ensures that the system not only delivers reliable performance but also creates an engaging, curriculum-aligned, and culturally localized learning platform.

3.3 Functional Modules & Workflow

The Hearing Impairment Support module will be structured into the following functional parts to ensure accessibility, engagement, and curriculum alignment:

1. User Registration & Login - Students register by providing details such as name, grade, and type of disability. Login options include password or Face ID for convenience.
2. Subject & Lesson Selection - Students select subjects and lessons aligned with the Sri Lankan Grade 3–5 curriculum.
3. Video Lessons with SLSL & Subtitles are delivered through videos with Sinhala Sign Language (SLSL) interpretation and clear Sinhala subtitles.
4. Text & Visual Lessons - Lessons are also available in text form with diagrams and images to enhance understanding.

5. Sign Language Practice - Students practice signs using the device camera. Pose estimation (MediaPipe + BlazePose) detects gestures, and AI models (GRU/1D-CNN + DTW) provide numeric scores (0–100) with real-time feedback.
6. Personalized Calibration - The system adapts to each learner’s signing style, ensuring fair and accurate scoring.
7. Sign Imitation Games - Gamified activities where students earn points, badges, and trophies for accurate signs, boosting motivation.
8. Animated Story-based Explanations - Complex subjects are explained with interactive animations and culturally familiar storylines.
9. AI Chatbot & Quizzes - A Sinhala-enabled chatbot answers questions and generates AI-based quizzes after each lesson, providing instant feedback and progress scores.
10. Progress Tracking - Parents and teachers can access progress reports showing completed lessons, quiz scores, and improvement areas.
11. Accessibility Features - Child-friendly design with large buttons, simple navigation, visual alerts, and vibration-based feedback to ensure inclusivity.
12. Data Security & Ethics - All student data is securely stored with anonymization and parental consent, ensuring compliance with ethical standards.

3.4 Data Collection

To ensure the application meets the real needs of hearing-impaired students, data will be collected from multiple sources, including the School of the Deaf in Tangalle and the National Institute of Education (NIE), Maharagama.

During the visit to NIE Maharagama, discussions were held with the Head of the Inclusive Education Section, who provided key insights on different types of students with disabilities and their learning methods. Specific guidance was given on dataset collection,

emphasizing that each sign must be recorded accurately with its corresponding landmarks. It was highlighted that even minor errors could alter the meaning of a sign, so recordings must be 100% precise. The NIE also provided explanations and access to image-based resources that support dataset creation.

At the School of the Deaf in Tangalle, real-world observations will be conducted to understand how students currently learn and the barriers they face when using traditional materials. Teacher interviews will provide valuable input on lesson delivery, communication challenges, and suggestions for creating effective learning modules. For building SLSL lessons and gesture recognition, certified interpreters will assist in recording signs, which will be used both for video-based lessons and AI model training.

Additionally, curriculum analysis will be carried out by reviewing the Sri Lankan national syllabus to ensure that all lesson content within the application is aligned with school standards. By combining expert consultation, direct classroom observation, interpreter-led recordings, and curriculum mapping, the dataset will remain accurate, culturally relevant, and educationally practical for both students and teachers.

3.5 Expected Outcomes

At the end of the project, the system will:

- Provide complete lesson access through SLSL videos, text, and diagrams.
- Allow interactive sign language practice with AI—based scoring.
- Include animated stories to explain complex topics.
- Offer AI—generated quizzes for self—assessment.
- Enable parents and teachers to track progress in real-time.
- Support cross-platform use on Android and iOS.

This methodology ensures that the application is practical, culturally relevant, and technically sound, giving hearing-impaired students in Sri Lanka an equal chance to learn and succeed in their studies.

3.6 PROJECT REQUIREMENTS

3.6.1 Functional Requirements

1. Lesson Delivery

- System shall display Sinhala Sign Language (SLSL) lessons in video format with clear subtitles.
- System should provide lesson content in text format with supporting diagrams and images.

2. Sign Recognition & Scoring

- System shall capture and analyze student hand/body gestures via the device camera.
- System shall detect signs using pose estimation frameworks (e.g., MediaPipe + BlazePose).
- System shall compare student gestures with reference signs and provide numeric scoring with real-time feedback.

3. Interactive Games

- System shall deliver lesson-based sign imitation games aligned with the Sri Lankan Grade 3–5 SLSL curriculum.
- System should provide real-time interactive feedback during games to enhance learning engagement.

4. AI Chatbot

- System shall allow students to ask questions in text form.

5. System should provide explanations, guidance, and instructions in Sinhala.

6. Quiz Generation

- System shall generate AI-powered quizzes automatically based on completed lessons.

- System shall display quiz questions and answers in text format and give instant feedback.

6. Progress Tracking

- System shall record and display completed lessons, quiz scores, mistakes, and gesture performance.
- System shall allow parents and teachers to log in and view progress reports through a dashboard.

7. User Authentication

- System shall support password-based and Face ID login options.
- System should maintain secure session handling and protect child data.

8. Backend Data Management

- System should store student progress, gesture scores, and reports in a secure backend using Firebase + MongoDB.
- System should manage large video lessons, AI models, and reports using cloud storage (AWS S3).

9. Mobile Accessibility

- System should operate on mobile devices (Android & iOS) through a cross-platform Flutter app.
- System shall ensure a child-friendly interface with simple navigation, visuals, and accessibility features.

3.6.2 Non-Functional Requirements

1. Performance
2. Accessibility
3. Usability
4. Security
5. Compatibility
6. Reliability
7. Maintainability

3.7 User Requirements

The system is designed to meet the needs of four main user groups. Students in Grades 3–5 require lessons, interactive games, and quizzes presented in SLSL and simple Sinhala text so they can understand without relying on sound. Parents need access to clear progress reports to monitor their child’s learning and development. Teachers require tools to track student performance, identify difficulties, and suggest improvements. Administrators need features to upload new lessons, manage quizzes, and monitor overall system usage to ensure smooth functioning.

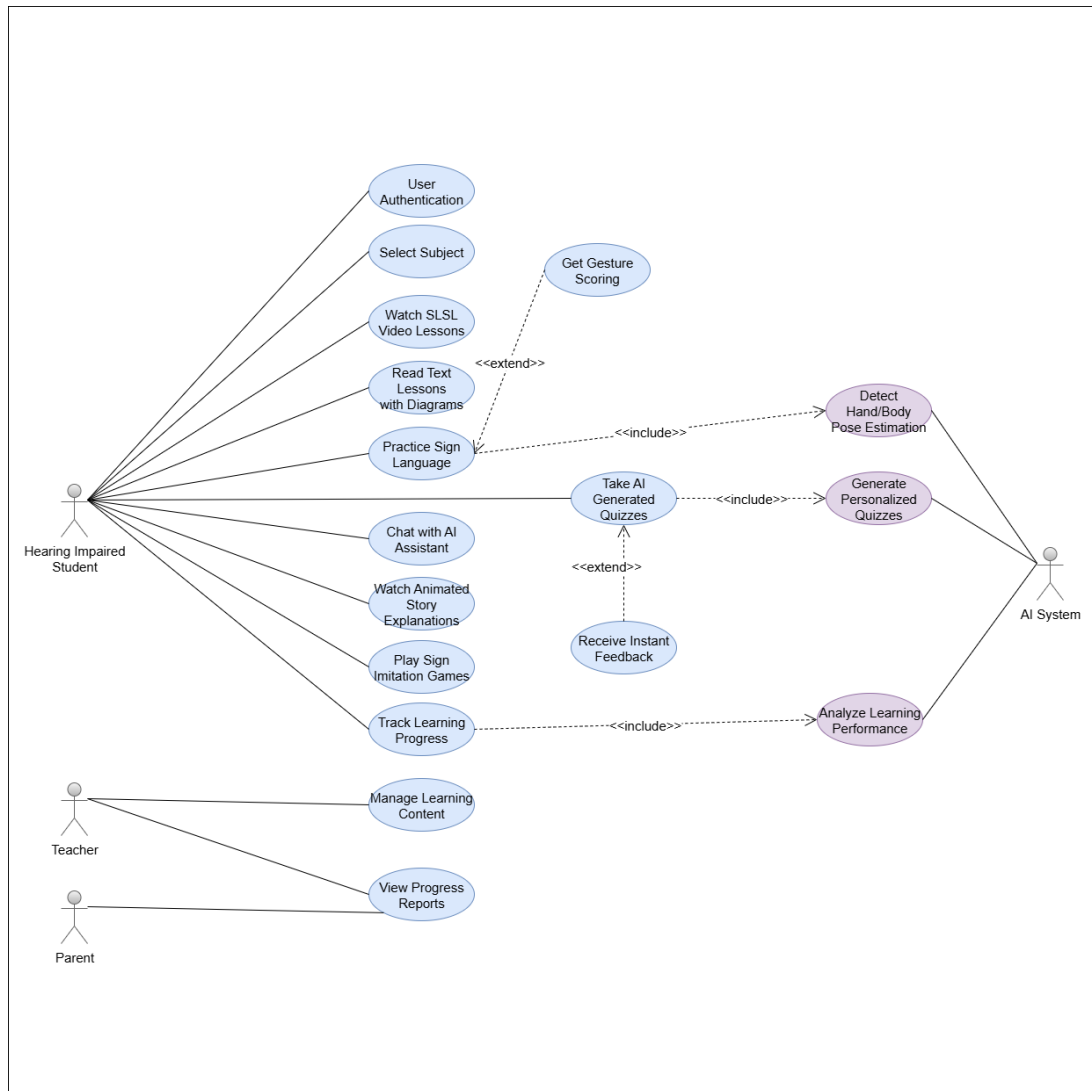


Figure 3: Use case diagram for hearing impairment support

4. BUDGET AND BUDGET JUSTIFICATION

Table II. Explains estimated costs for tools and resources to assist hearing-impaired students.

Table II.: Budget Plan for Inclusive Mobile Learning App for hearing impairment support

Purpose	Estimate Cost (LKR)
Travel to special schools, interviews with teachers & students	6000
Recording lessons with interpreter support	10000
Internet cost	8000
Documentation & Printing	1500
Total	25500

5. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The commercialization of the Hearing Impairment Support application focuses on addressing a clear market gap in Sri Lanka's education sector. Currently, there is no mobile application that provides curriculum-aligned lessons in SLSL for primary school students in Grades 3–5. By integrating sign imitation games, gesture scoring and AI-driven quizzes, the product stands out as a unique and innovative solution tailored to the needs of local learners. This differentiation establishes a strong foundation for commercialization by offering both novelty and practical value.

The primary market for this product includes approximately 2,400 deaf students in Grades 3–5 across Sri Lankan government schools, along with a broader community of hearing-impaired learners in special education centers. In addition, the solution can be extended to teacher training institutions, inclusive schools.

The product also has strong entrepreneurial potential through collaborations with private educational organizations and EdTech companies. Subscription-based models, school licensing packages, and partnerships with telecom providers for subsidized access can ensure sustainable revenue generation. Furthermore, the product can be adapted into a

freemium model, where basic lessons are offered for free while advanced sign-imitation games, analytics, and progress tracking features are monetized.

The product's unique selling points include curriculum alignment, real-time gesture scoring, gamified learning features, and localized SLSL content. These make it highly attractive to schools, special education centers, and parents seeking structured digital tools for their children. With growing government focus on inclusive education, the Hearing Impairment Support application holds strong potential for sustainable commercialization and impactful entrepreneurship in the EdTech sector.

6. WORK BREAKDOWN STRUCTURE

Task \ Time	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
Requirement analysis & Research								
Data gathering, Proposal Doc & Presentation								
Prototype Design								
App Development								
AI related Developments								
Integration & Testing								
Validation & Feedback								
Final Presentation								

Figure 4: Work breakdown structure for Shilpa mobile app

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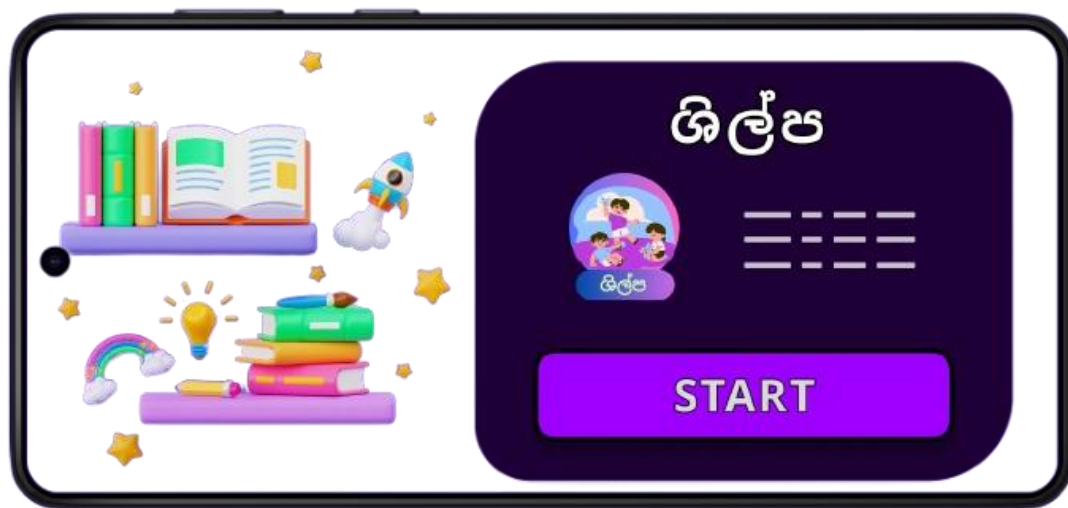
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8. APPENDICES

Appendix A: Inclusive Mobile App logo (Shilpa)



Appendix B: Application Loading Page Interface



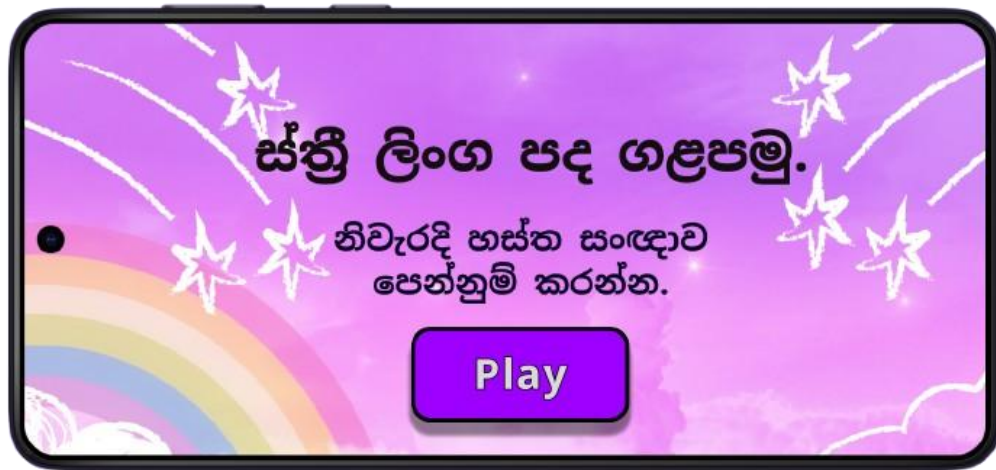
Appendix C: User Interface for Disability Type Selection



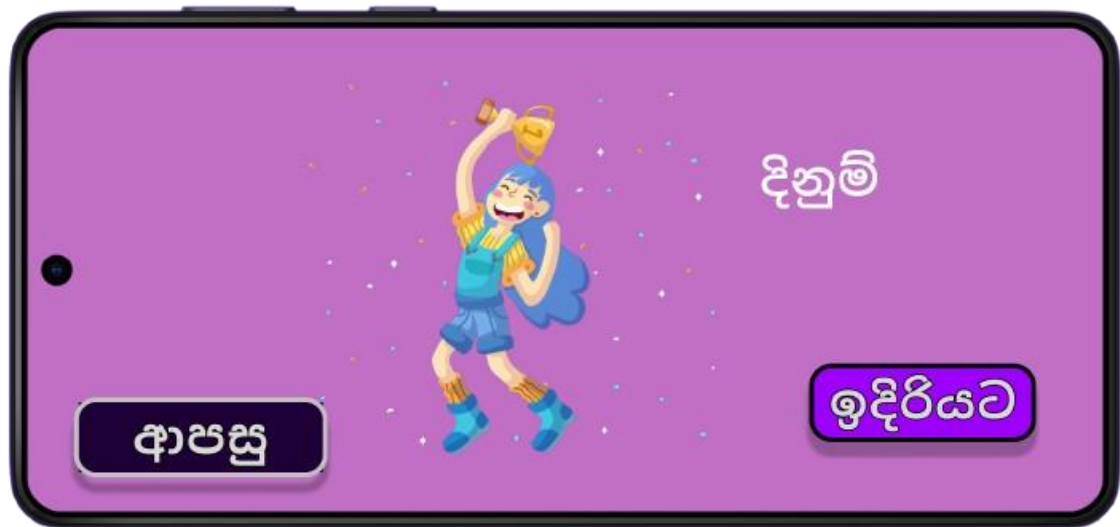
Appendix D: Lesson Delivery Support for Hearing-Impaired Students



Appendix E: Sign Imitation–Based Educational Games



Appendix F: Game Completion and Reward Display



Appendix G: Data Collection Visit at the National Institute of Education (NIE)



- Appendix H: Plagiarism Report



Class Portfolio

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